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MIL-R-85111A(AS)
AMENDMENT 8
15 September 1999
SUPERSEDING
AMENDMENT 7
26 October 1992

MILITARY SPECIFICATION

RECOVERY ASSIST, SECURING AND TRAVERSING SYSTEM FOR LAMPS MK III HELICOPTER

This amendment forms a part of MIL-R-85111A(AS), dated 10 December 1980, and is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1 SPECIFICATIONS, FEDERAL: Following "RR-W-410 listing, add:

"TT-C-490 Cleaning Method and Pretreatment of Ferrous Surfaces for Organic Coatings"

* 2.1 SPECIFICATIONS, FEDERAL: For the TT-P-645 listing, replace "chromate" with "molybdate."

2.1 SPECIFICATIONS, MILITARY: Delete "MIL-C-915 General Specification for Electrical Cable and Cord for Shipboard Use."

PAGE 2

2.1 SPECIFICATIONS, MILITARY: Following "DOD-P-17545" listing, add:

"MIL-H-17672 Hydraulic Fluid, Petroleum, Inhibited"

2.1 SPECIFICATIONS, MILITARY: Following "DOD-P-24555" listing, add:

"MIL-C-24643 Cable, Electrical, 600 Volts"

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* 2.1 SPECIFICATIONS, MILITARY:

Delete DOD-P-17545 Primer Coating, Alkyd Red Lead Type, Formula No. 116 and Formula No. 116D."

PAGE 3

2.1 STANDARDS, MILITARY: Following "MIL-STD-831" listing, add:

"MIL-STD-869 Flame Spraying"

PAGE 5

3.1.1.2, third sentence: Delete in its entirety.

3.2.2.1, line 5: Delete "MIL-H-83282" and substitute "MIL-H-17672."

PAGE 6

3.2.2.4, line 11: Delete "MIL-H-83282" and substitute "MIL-H-17672".

PAGE 7

3.2.2.6, line 10: Change "150 lbs" to "180 lb."

3.2.2.7: Delete and substitute:

"3.2.2.7 Control console. The control console shall contain all of the system controls, indicators, and switches to be employed in the normal operation of the RAST system. The console shall conform to the installation and interface requirements of Installation Control Drawing NAEC 700,000, except that provisions for operator voice communications and the connection and control thereof are not required. The console, when combined with other units of the RAST system into a system configuration, shall conform to the system performance requirements of 3.4.1."

3.2.2.8: Delete in its entirety.

PAGE 11

3.4.1.2a, line 3: Delete "In either the Tail Probe Grid or."
3.4.1.2a, line 14: Delete "grid or".

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3.4.1.3a, line 3: Delete “and the Tail Probe engaged in the Tail Probe Grid.”

3.4.1.3d: Delete and substitute:

“d. A capability to reel-out the tailguide cables shall be provided in which the steady state force required at the on-deck ends to unspool the cables does not exceed 75 lb.”

PAGE 13

3.4.1.3g, line 7: Delete 0.2 g” and substitute “0.3g.”

PAGE 15

* 3.4.2.2.2.5: Change the name of Formula No. 84 from “Zinc chromate primer” to “Zinc molybdate primer.”

PAGE 16

* 3.4.2.2.2.5: Delete from the list:

Formula No. 116 Red lead primer, alkyd DOD-P-17545

* 3.4.2.2.2.5.1, subparagraph a: Delete “or Formula 116” in the first sentence.

* 3.4.2.2.2.5.1, subparagraph b: Delete “or 116” from the first sentence.

* 3.4.2.2.2.5.1, subparagraph c: Delete the last sentence of the paragraph and substitute:

“Such parts shall then be given one coat of epoxy primer Formula 150 and finished with one coat of Formula 111, Class 2.”

* 3.4.2.2.2.5.2, subparagraph a: Delete the words “or 116” from the first sentence.

3.4.2.2.2.5.1, add subparagraph e:

“e. For those surfaces normally subjected to abrasive wear, a two-layer plating system shall be required. This plating system shall consist of an electroless nickel plate and an electrolytic-electroless nickel plate. The following components shall be coated with a two-layer nickel plating system: steel sheaves (above and below deck) including rope accumulator sheaves and tachometer sheaves, drums of the movable sheave gear box, Recovery Assist Winch of the Winch Hydraulic Power Unit (WHPU), Tail Guide Winch (TGW) and traverse winch, and the sheave follower on the Recovery Assist Winch (WHPU).”

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3.4.2.2.2.5.1, add subparagraph f:

"The tail guide exit sheave assemblies, down sheave assemblies, and traverse track sheave assemblies shall have all manufactured steel components (except sheaves) that are not welded to the deck, coated with a single-layer electroless nickel plating."

PAGE 18

3.4.2.2.2.5.6a: Delete the first line and substitute the following:

"a. Steel. Unless coated in accordance with 3.4.2.2.2.6 or nickel plated in accordance with 3.4.2.2.2.5.1, steel surfaces …."

3.4.2.2.2.5.6a, line 3: Delete "track trough components and tail guide sheave well components."

3.4.2.2.2.5.6, add subparagraph c:

"c. Arresting Beams, Rapid Securing Device. The beams shall be galvanically protected with 0.005 inch minimum thickness of aluminum flame spray in accordance with MIL-STD-869. The topcoat shall consist of 0.004 to 0.006 inch dry film thickness of primer followed by 0.003 to 0.005 inch fused film thickness of Nylon 11."

PAGE 19

3.4.2.2.2.6c(4), delete the last sentence and substitute the following:

"The external surface of the track bolt inserts, when utilized, shall be treated with the ceramic (SermeTel or equal) coating."

3.4.2.2.2.6: Add:

"d. Epoxy powder coatings. Consideration shall be given to the use of epoxy powder coatings for exposed ferrous parts subject to abrasion and impact. Surface preparation for powder coatings shall be abrasive blasting in accordance with 3.4.2.2.2.3a, with the addition of iron phosphating in accordance with TT-C-490, Type II. Dry film thickness shall be as follows:

Minimum:	3 mils
Maximum:	10 mils (average of 4 measurements. No single measurement greater than 15 mils.)

If required, a finish coat to match the surrounding structure may be applied."

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3.4.2.2.2.6, add subparagraph e:

"e. Two-layer plating system. The nickel plate wear-resistant plating consists of the sequential application of an electrolytic-electroless nickel two-layer plating. Following these applications, the two-layer plating shall be subjected to final heat treatment to establish the hardening of the surface."

PAGE 21

3.4.5.3: Delete in its entirety.

3.4.5.5: Delete in its entirety.

3.4.5.6, lines 3 and 4: Delete "and the enclosure of the Control Console."

PAGE 22

3.5.5.2, line 3: Delete "MIL-C-915" and substitute "MIL-C-24643."

Add last sentence: "Connectors that are exposed to the environment before shipboard installation shall have shipping closures in accordance with MS90376 protective plastic dust caps."

3.5.6: Delete the last sentence and substitute:

"Couplings that are exposed to the environment before shipboard installation shall have aluminum alloy closures furnished with a dust and moisture sealing gasket or seal resistant to MIL-H-17672 hydraulic fluid."

PAGE 27

TABLE II, Subject column, line 14: Delete "Full."

TABLE II, line 23: Delete "Salt fog, 3.4.5.3, 4.8.3.6, X."

TABLE II, line 27: Delete "Dust, 3.4.5.5, 4.8.3.10, X."

PAGE 29

4.8.2.1: Delete and substitute:

"4.8.2.1 System performance test. System equipment, installed in accordance with established installation and interference requirements and combined into a representative system configuration, shall be subjected to a performance test to determine compliance with 3.4.1. The

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representative system configuration shall include at least one each of the individual items described in 3.2.2 and include the capability to demonstrate the performance of each system functional interface, internal and external; at least one track incorporating both straight and curved sections shall be employed. The test shall be performed as approved in accordance with 4.3."

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TABLE III, line 10: Delete "4.8.3.6, Salt Fog" and "RSD, CC, ECR, IB for ECR."

TABLE III, line 14: Delete "4.8.3.10 Dust" and "CC, ECR, RSD."

4.8.3.2, line 5: Delete "and Control Console are" and substitute "is."

4.8.3.6: Delete.

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4.8.3.10: Delete.

NOTE: The margins of this amendment are marked with asterisks to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Preparing activity:
Navy – AS

(Project 1710-0065)

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